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Case Study: Jaela Maxcream Private Limited (CreamRo)

Course Code: IS4108 — Information Systems Strategy
Institution: Faculty of Computing, Sabaragamuwa University of Sri Lanka
Group: Group 17 Submission
Date of Submission: May 30, 2026

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1. Executive Summary

This comprehensive study delivers an exhaustive, critical analysis of the strategic orchestration of Information Systems (IS) and automated technology architectures within the contemporary Sri Lankan Food and Beverage sector, focusing directly on the operational realities of Jaela Maxcream Private Limited (operating under the commercial consumer brand name *CreamRo*). As an emerging, agile fast-moving consumer goods (FMCG) startup based in the industrial hub of Ekala, the enterprise functions inside a highly concentrated and aggressively hostile competitive landscape. This environment is structurally dominated by historical, multi-billion rupee market incumbents, most notably Elephant House (Ceylon Cold Stores PLC) and Cargills Ceylon PLC. Rather than deploying technology through a short-sighted lens focused exclusively on immediate transactional profit maximization, Maxcream’s overarching IS strategy is fundamentally anchored on achieving architectural efficiency, absolute biological cleanliness, and uncompromising raw material quality. This report deconstructs and evaluates their complete operational layout, analyzing sensor-driven raw milk intake networks, fully automated computer-controlled fluid routing lines, real-time volumetric packaging stabilization loops, and advanced thermodynamics controls. It further maps their protective risk-mitigation tactics designed to prevent corporate recipe reverse-engineering, and sets forth a future digital roadmap detailing Internet-of-Things (IoT) reefer fleet telematics and retail-end capacity forecasting systems necessary to scale and sustain their market positioning.

2. Introduction & Organizational Profile

The food and beverage industry—particularly the dairy processing and ice cream manufacturing sub-sector within Sri Lanka—represents one of the most volatile, logistically demanding, and capital-intensive domains within the broader fast-moving consumer goods (FMCG) market. Success within this space is rigidly governed by a combination of tight national nutritional standard compliance, fast speed-to-market execution, and seamless, uninterrupted cold-chain logistics preservation. Within this challenging ecosystem, Jaela Maxcream Private Limited operates from its centralized, specialized manufacturing facility situated in the industrial zone of Ekala. As a highly responsive manufacturing startup, Maxcream produces mass-market dairy confections under the commercial brand name *CreamRo*. Its product portfolio ranges from high-volume bulk items, such as its four-liter family packs, down to single-serve impulse lines tailored for supermarkets, commercial food-service traders, and localized retail networks. The fundamental challenge defining Maxcream's operational existence is the mature, entrenched nature of its domestic market environment. The Sri Lankan ice cream industry is duopolistically controlled by large conglomerates that possess deep

capital reserves, extensive multi-tier distribution networks, and massive economies of scale. For a newly launched startup like Maxcream, competing on capital expenditure or price wars is a strategic impossibility. Consequently, the company has deployed a targeted information systems strategy that substitutes raw corporate scale with high operational agility. By designing an automated, high-throughput production plant that runs with minimum operational friction and minimal manpower costs, Maxcream successfully sustains a lean, highly competitive market profile.

3. Research Methodology & Primary Data Collection

To ensure empirical validity, this study combines a comprehensive secondary review of domestic dairy supply chain mechanics with a rigorous primary data collection framework. Primary qualitative data was obtained via an immersive, on-site field observation and an extensive, structured technical interview conducted with Maxcream's corporate Production Executive at the Ekala manufacturing plant.

The field research focused on mapping exact data flows, systemic command sequences, and physical material transformations across the factory floor. This mapping began at the raw milk inbound receiving bays and extended through processing vats, continuous freezing tunnels, packaging lines, flash-hardening cells, and deep cold-room storage vaults.

The resulting interview transcript was qualitatively deconstructed utilizing a socio-technical analytical approach. This process systematically categorized the enterprise's software dependencies, hardware sensor loops, manual operational boundaries, competitive vulnerabilities, and future digital investment roadmaps. This methodology bridges theoretical information systems frameworks with the practical realities of a functioning manufacturing floor.

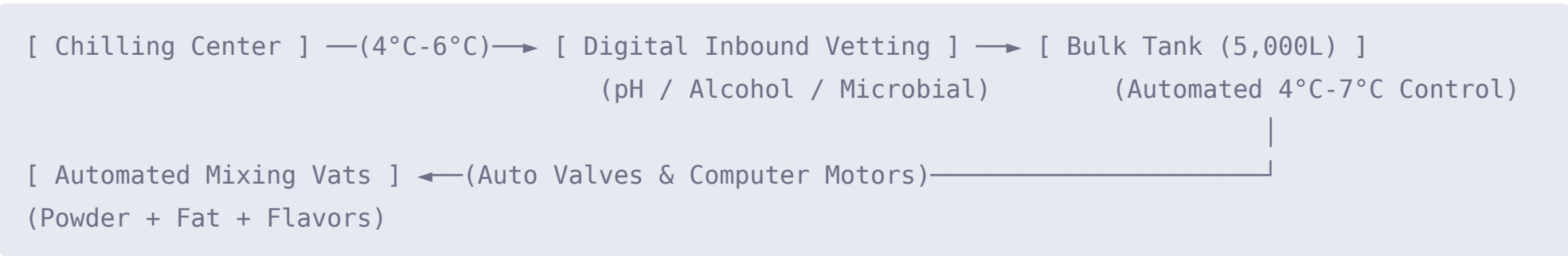
4. Strategic Alignment & Competitive Positioning Analysis

Maxcream's deployment of digital tools presents a powerful application of Michael Porter's Generic Strategies, specifically adapting the cost leadership and focus differentiation paradigms to an agile startup framework. While conventional enterprise technology frameworks in consumer goods are typically engineered around front-end consumer data mining, predictive demand modeling, and price optimization, Maxcream strategically applies its IS infrastructure directly to the back-end production architecture to maximize asset productivity and eliminate waste.

According to corporate leadership, the deployment of human labor within the core processing chain introduces unacceptable cost structures, operational delays, and significant biological contamination risks. For example, validating twenty distinct inbound raw milk samples manually using traditional laboratory methods is completely impractical within high-volume manufacturing environments due to the severe time constraints imposed by perishable inputs. By implementing automated information control loops, Maxcream swaps slow, manual labor for rapid, reliable automated processing. This strategic alignment optimizes factory utilization, guarantees strict compliance with biological safety standards, minimizes material waste, and allows the enterprise to achieve an ultra-lean cost structure capable of defending against aggressive market incumbents.

5. Core Operational Information Systems & Production Pipeline

The operational layout of Maxcream's Ekala facility is structured into four highly coordinated technological pathways that bridge physical machinery with digital logic networks: inbound supply chain collection, automated batch processing, continuous freezing/filling, and thermodynamic flash-hardening.



5.1 Inbound Supply Chain Architecture & Sensor Data Ingestion

The integrity of the dairy cold chain begins at external regional chilling centers where raw fresh milk is gathered from local agricultural networks. Because raw milk is highly perishable, it must be cooled immediately and maintained at an optimum transit temperature between 4°C and 6°C to prevent microbial multiplication and souring. Upon arrival at the Ekala plant's receiving bays, the inbound raw milk immediately undergoes automated quality vetting. Rather than relying on slower manual chemistry protocols, the factory utilizes digital sensor arrays and electric pH meters.

These digital tools provide immediate, accurate readings of critical quality indicators, including raw pH levels, alcohol

concentration fractions, and underlying microbial components. Manual testing methods are avoided at this stage because they fail to deliver the immediate, accurate readings required to support high-speed processing. Once the raw milk passes these automated quality checks, it is transferred into the central bulk milk tank. This bulk storage unit forms the core of the inbound logistics network, managing a maximum target throughput of up to 5,000 liters per day. The storage environment is continuously regulated by an automated control loop that coordinates temperature sensors with a digital control panel. If internal temperatures drift, the system automatically adjusts cooling cycles to keep the raw milk strictly within a safe storage range between 4°C and 7°C, preserving material stability before processing begins.

5.2 Automated Line Processing, Software Routing & Valve Controls

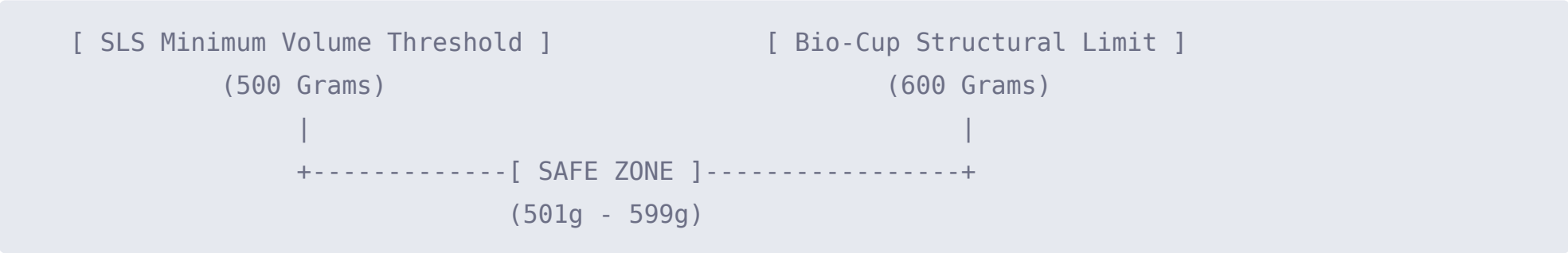
To prevent spoilage and lock in nutritional values, the factory processes fresh milk within one to two days of collection. The manufacturing routing lines connecting storage tanks to mixing units are fully automated, utilizing a localized industrial network to minimize manual handling and contamination risks. Plant operators interface with the line through central computer terminals, entering specific volume parameters and process metrics for each batch. Once these commands are executed, the central software sends digital instructions to automated valves and physical line motors via sensor networks. The valves open automatically, allowing the raw milk to flow under computer control into the main mixing vats. Continuous sensors measure liquid volume levels in real time; as soon as the fluid reaches the targeted parameter threshold, the software triggers the line valves to shut automatically. Within these processing tanks, the automated system manages the blending of raw milk powder, butter, fat solids, and specific flavor extracts, preparing the baseline mix for the subsequent freezing and packaging cycles.

6. Quality Assurance, Compliance, & SLS Standardization Infrastructure

For a growing food manufacturer in Sri Lanka, complying with national standardization benchmarks is a critical requirement for securing shelf space in premium modern trade supermarket networks like Arpico. Maxcream handles these regulatory pressures by embedding strict quality compliance parameters directly into the software tracking loops of its packaging and freezing machinery.

6.1 Volumetric Calibration and Bio-Cup Structural Safety Controls

The plant's continuous freezers handle both the freezing process and automated volume monitoring. Maxcream differentiates its brand packaging by using biodegradable paper-based container alternatives, known as bio-cups, rather than the standard plastic tubs favored by larger market incumbents like Cargills Ceylon and Elephant House. While sustainable, these paper-based bio-cups introduce specific physical and mechanical limits on the line.



Primary field data shows that a single bio-cup container has a strict maximum weight limit of 600 grams of ice cream. If a batch exceeds 600 grams, the physical joints of the container suffer from excessive mechanical strain, causing the paper walls to buckle, leak, or tear along the seams. However, to comply with the country's national SLS certification standards, the volume of a one-liter retail container must remain strictly above a minimum weight baseline of 500 grams. This volume-to-weight balance is monitored automatically by the continuous freezers, which continuously track and adjust filling outputs. This software-driven volume control ensures every unit meets SLS standards while remaining safely below the 600-gram limit to prevent packaging failures.

6.2 Fast-Hardening Automation and Cooling Room Cycles

Immediately following the filling stage, the packaged bio-cups are routed via automated lines into the specialized hardening unit. The setting phase is a critical step in ice cream manufacturing, as rapid cooling is essential to stabilize the product's cream texture and prevent the growth of large ice crystals. While the baseline ambient room temperature inside the Sri Lankan facility averages roughly 27°C, the automated hardener drops temperatures down to a severe -35°C, rapidly chilling the product to its solid-state setpoint. After passing through the hardener, the inventory is transferred into the main walk-in cooling rooms, where temperatures are

held at a constant -37°C. The cooling room runs complex, automated air-circulation and regeneration cycles. This software-driven airflow management system optimizes internal cooling efficiency while actively reducing heat generation, preventing localized warm spots that could compromise product quality or cause batch spoilage during storage.

7. Strategic Marketing & Risk Containment Frameworks

As an emerging startup balancing restricted capital with the need to establish market visibility against dominant giants, Jaela Maxcream adopts an agile, cost-effective digital marketing approach. The enterprise bypasses high-cost traditional media spaces, focusing instead on organic customer engagement across active corporate channels on Facebook, Instagram, and TikTok.

To maximize visibility on TikTok, the company collaborates with content creators and page holders to amplify brand awareness through targeted short-form video content. This lean approach allows the startup to market its brand and products within modern trade networks and general retail markets without incurring heavy traditional advertising expenses.

Despite having active digital channels, Maxcream’s overarching marketing strategy is intentionally controlled and bounded due to clear competitive risks. In the highly concentrated Sri Lankan dairy landscape, dominant market leaders like Elephant House and Cargills closely monitor the market for openings to crowd out new entrants. Management highlights that large competitors actively look for strategic or operational missteps to eliminate newcomers from retail shelves.

Historical analysis in the food and beverage industry shows that large-scale, aggressive marketing campaigns can provoke crushing retaliatory responses from established giants, similar to the historical market wars between Coca-Cola and Pepsi. For example, major brands can permanently shift consumer cultural perceptions through massive advertising initiatives, such as Coca-Cola's historic role in standardizing the modern red imagery of Santa Claus.

Because Maxcream cannot match the massive advertising budgets of its larger rivals, any misstep in their digital messaging could trigger aggressive pricing or distribution pushback from incumbents. Therefore, the marketing department deliberately maintains a controlled social media footprint, adjusting its approach over time based on steady growth and market shifts rather than pursuing high-risk, aggressive expansion.

The most critical competitive threat identified on the factory floor is the risk of intellectual property theft and product reverse-engineering. Modern competitors can use advanced laboratory testing to analyze physical ice cream samples, allowing them to reverse-engineer ingredient ratios, sugar content, flavor profiles, and coloring agents. To counter this risk and prevent digital corporate espionage, Maxcream implements a strict data isolation policy. The core product recipes and chemical ingredient formulations are kept completely offline and are never entered into any networked computer system or software database. Knowledge of the master recipe is restricted to a single individual within the company, creating an effective data firewall that removes the risk of external digital leaks or network breaches.

8. Challenges, Security Vulnerabilities, & System Constraints

A key operational finding from the field study is the clear gap between process automation and structural accuracy within food processing lines. While automated information systems excel at high-speed processing and volume regulation, they cannot evaluate complex quality variables within specialized product lines. This creates an operational paradox where human supervision remains essential for accuracy.

This limitation is clearly demonstrated in the manufacturing of specialty items, such as the premium fruit-and-nut product line. Current automated sensors can only calculate the total weight of a filled container; they cannot distinguish between the base ice cream mix and secondary mix-ins like fruit chunks.

To comply with internal quality standards, a standard 500ml specialty tub must contain exactly 115 grams of fruit pieces. Because automated systems cannot isolate and measure these components on the line, the process requires manual intervention. Production executives must manually weigh and audit ingredient ratios to ensure product consistency, illustrating a key limitation where human oversight is required to bridge the gaps in automated manufacturing systems.

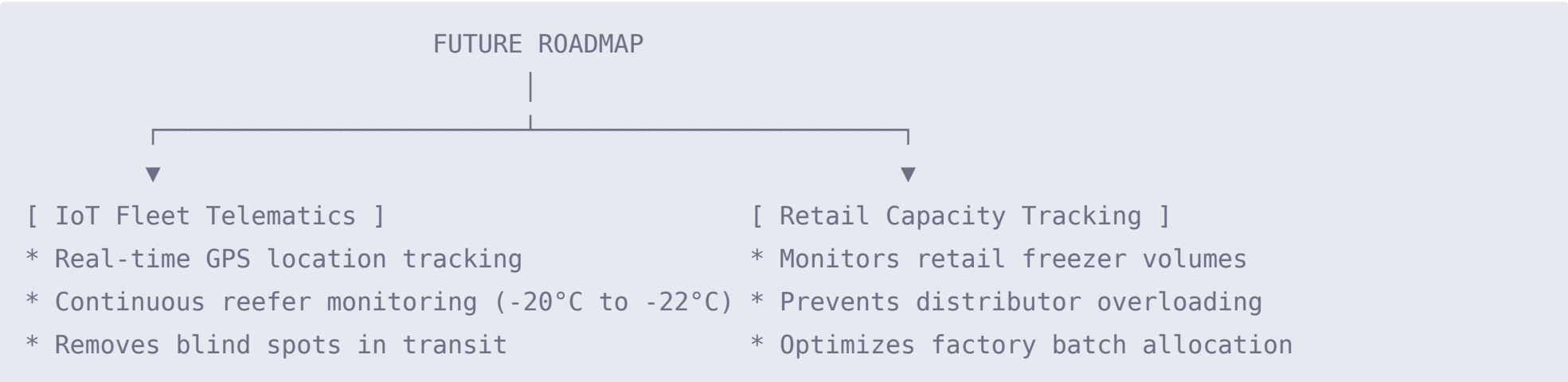
THE AUTOMATION-ACCURACY PARADOX	
Automated Continuous Freezers	Human Quality Executive
* High-speed volumetric throughput	* Manual ingredient audits
* Monitors total container weight	* Checks complex item mixes

| * Cannot separate sub-component weights | * Measures exact fruit weight |

As Maxcream updates its production plant with advanced automated networks and automated routing valves, it faces typical socio-technical challenges regarding labor displacement. Currently, the automated ingredient blending systems and sensor networks run with minimal manual intervention, significantly reducing plant labor requirements. While this automation reduces human error and lowers operational costs, management acknowledges the long-term risk of worker displacement within the local manufacturing plant. Consequently, future technology integrations must carefully balance automation-driven productivity gains with sustainable human resource management.

9. Future IS Strategy Roadmap: Cold-Chain IoT & Smart Logistics

To expand its market reach across modern trade networks, Maxcream is planning two core updates to its information systems strategy to eliminate structural visibility blind spots.



9.1 IoT Fleet Telematics Network Integration

While Maxcream’s internal factory floor is highly automated, its outbound logistics network currently faces clear data visibility gaps. Outbound transportation relies on refrigerated delivery trucks (reefers) to move finished products to retail centers and regional distributors. These trucks are designed to maintain internal temperatures between -20°C and -22°C, but they currently lack real-time centralized tracking. This leaves management blind to temperature fluctuations, delivery delays, or cold-chain breaks during transit.

To resolve this limitation, Maxcream is collaborating with an external technology provider to deploy an integrated Internet-of-Things (IoT) fleet telematics platform. The upcoming system will integrate GPS modules and temperature sensors into each delivery vehicle, continuously streaming location and climate data back to a central management dashboard. This infrastructure will remove visibility gaps in logistics, allowing management to actively monitor cold-chain integrity and verify delivery conditions until the product reaches the buyer.

9.2 Retail Endpoint Capacity Optimization Systems

The second major initiative in the company's future roadmap addresses demand distortion and distributor inventory issues. Sales teams frequently run into issues where retail distributors request additional inventory despite having overloaded freezers. Overloading retail freezers blocks proper airflow, causing temperature fluctuations that degrade ice cream texture and damage packaging.

To address this challenge, Maxcream plans to implement a retail endpoint inventory tracking system. This application will allow sales teams to track and log retail freezer storage levels before dispatching shipments. By monitoring available endpoint capacities, Maxcream can optimize delivery schedules, prevent inventory losses from poor retail storage, and efficiently allocate production batches to match real market demand.

II. References (IEEE Format)

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